

Chapter 3

Exercises

Group 2

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Exercise 1 - Problem Statement

Question

We are given the following corpus, modified from the one in the chapter:

```
<s> I am Sam </s>  
<s> Sam I am </s>  
<s> I am Sam </s>  
<s> I do not like green eggs and Sam </s>
```

Using a bigram language model with add-one smoothing, what is $P(\mathbf{Sam} \mid \mathbf{am})$? Include `<s>` and `</s>` in your counts just like any other token.

1. The Add-One (Laplace) Smoothing Formula

To find the smoothed bigram probability $P(w_n|w_{n-1})$, we use:

$$P_{Laplace}(w_n|w_{n-1}) = \frac{C(w_{n-1}, w_n) + 1}{C(w_{n-1}) + V}$$

- $C(w_{n-1}, w_n)$: Count of the bigram in the corpus.
- $C(w_{n-1})$: Count of the unigram (the history).
- V : Total number of unique word types in the vocabulary.

Exercise 1 - Solution

2. Counting from the Corpus

Vocabulary $V = \{ \langle s \rangle, I, am, Sam, \langle /s \rangle, do, not, like, green, eggs, and \}$
 $\rightarrow V = 11$

Count of history $C(am)$:

- $\langle s \rangle I am Sam \langle /s \rangle (1)$
- $\langle s \rangle Sam I am \langle /s \rangle (1)$
- $\langle s \rangle I am Sam \langle /s \rangle (1)$

$\rightarrow C(am) = 3$

Count of bigram $C(am Sam)$:

- $\langle s \rangle I am Sam \langle /s \rangle (1)$
- $\langle s \rangle I am Sam \langle /s \rangle (1)$

$\rightarrow C(am Sam) = 2$

Exercise 1 - Solution

3. Final Calculation

Substitute the extracted values into the Laplace formula:

$$P(\text{Sam} \mid \text{am}) = \frac{C(\text{am Sam}) + 1}{C(\text{am}) + V}$$

$$P(\text{Sam} \mid \text{am}) = \frac{2 + 1}{3 + 11}$$

$$P(\text{Sam} \mid \text{am}) = \frac{3}{14} \approx 0.214$$

Conclusion

With Add-1 smoothing, the probability $P(\text{Sam} \mid \text{am})$ is **3/14** (or $\sim 21.4\%$).

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Exercise 2 – Problem Statement

Corpus

<s> I am Sam </s>

<s> Sam I am </s>

<s> I am Sam </s>

<s> I do not like green eggs and Sam </s>

Task

Using linear interpolation between a bigram MLE and unigram MLE model:

$$\hat{P}(w_n | w_{n-1}) = \lambda_1 P_{ML}(w_n | w_{n-1}) + \lambda_2 P_{ML}(w_n)$$

with $\lambda_1 = \lambda_2 = \frac{1}{2}$, compute: $P(\text{Sam} | \text{am})$

Include <s> and </s> in all counts.

Exercise 2 - Solution

Linear Interpolation Model

The interpolated probability is defined as:

$$\hat{P}(w_n|w_{n-1}) = \lambda_1 P_{ML}(w_n|w_{n-1}) + \lambda_2 P_{ML}(w_n)$$

where:

$$\lambda_1 = \frac{1}{2}, \quad \lambda_2 = \frac{1}{2}, \quad \lambda_1 + \lambda_2 = 1$$

Target Quantity

$$\hat{P}(\text{Sam}|\text{am}) = \frac{1}{2} P_{ML}(\text{Sam}|\text{am}) + \frac{1}{2} P_{ML}(\text{Sam})$$

Exercise 2 - Solution

Definition

The MLE bigram probability is:

$$P_{ML}(w_n|w_{n-1}) = \frac{C(w_{n-1}, w_n)}{C(w_{n-1})}$$

Counting from the Corpus

Occurrences of am:

$$C(\text{am}) = 3$$

Occurrences of bigram (am Sam):

$$C(\text{am Sam}) = 2$$

MLE Bigram Probability

$$P_{ML}(\text{Sam}|\text{am}) = \frac{C(\text{am, Sam})}{C(\text{am})} = \frac{2}{3}$$

Exercise 2 - Solution

Definition

$$P_{ML}(w) = \frac{C(w)}{N}$$

where N is the total number of tokens.

Total Token Count

Sentence lengths (including $\langle s \rangle$ and $\langle /s \rangle$):

$$N = 5 + 5 + 5 + 10 = 25$$

MLE Unigram Probability

$$P_{ML}(\text{Sam}) = \frac{4}{25}$$

Exercise 2 - Solution

Substitute into Interpolation Formula

$$P(\text{Sam}|\text{am}) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{4}{25}$$

Algebraic Simplification

$$P(\text{Sam}|\text{am}) = \frac{1}{3} + \frac{2}{25}$$

Common denominator:

$$P(\text{Sam}|\text{am}) = \frac{25}{75} + \frac{6}{75} = \frac{31}{75} \approx 0.413$$

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Exercise 3 - Problem Statement

Question

You are given a training set of 100 numbers that consists of 91 zeros and 1 each of the other digits 1-9.

Now we see the following test set:

0 0 0 0 0 3 0 0 0 0

What is the **unigram perplexity**?

Exercise 3 - Solution

1. Extract Unigram Probabilities (From Training Set)

The training set has $N_{train} = 100$ tokens.

- Count of '0' = 91 $\rightarrow P(0) = \frac{91}{100} = 0.91$
- Count of '3' = 1 $\rightarrow P(3) = \frac{1}{100} = 0.01$

2. Calculate Probability of the Test Set

The test set W has $N_{test} = 10$ tokens: nine '0's and one '3'. Assuming the unigram model (independent probabilities):

$$P(W) = P(0) \times P(0) \times P(0) \times P(0) \times P(0) \times P(3) \times \dots$$

$$P(W) = P(0)^9 \times P(3)$$

$$P(W) = (0.91)^9 \times (0.01)$$

Exercise 3 - Solution

3. Apply the Perplexity Formula

Perplexity is the inverse probability of the test set, normalized by the number of words N :

$$\text{Perplexity}(W) = \sqrt[N]{\frac{1}{P(w_1 \dots w_N)}}$$

Substitute our values ($N = 10$):

$$\text{Perplexity}(W) = \sqrt[10]{\frac{1}{(0.91)^9 \times 0.01}}$$

Final Result

Calculating the root yields:

$$\text{Perplexity} \approx 1.72$$